

Transfer life prediction of gears by cross-domain health indicator construction and multi-hierarchical long-term memory augmented network

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ABSTRACT

The long-term remaining useful life (RUL) prediction of gears is crucial for the safe operation and maintenance of rotating machinery. However, most existing RUL prediction methods face great challenge under the variable working conditions due to the lack of enough prior run-to-failure data. Therefore, this paper addresses to explore a new transfer life prediction methodology for gears. A gear health indicator (HI) transfer construction framework named TQFMDCAE is first proposed by a quadratic function-based multi-scale deep convolutional auto-encoder and maximum mean discrepancy, and it can generate the cross-domain HIs under different working conditions. Next, a novel RNN-based network named multi-hierarchical long-term memory augmented network (MLMA-Net) is developed for the life prediction of gears based on the obtained HIs. In MLMA-Net, a new memory augmentation function is intended to increase the network's long-term memory capacity. The proposed multi-hierarchical mechanism then divides the sequence information of the network into three attention hierarchies and three cell hierarchies, respectively. Experiments on equipment indicate that the developed MLMA-Net has a remarkable predictive capacity, particularly for predicting the long-term life of an object. Meanwhile, comparative results demonstrate that the proposed RUL prediction methodology is superior to other typical RUL estimation methods.

1. Introduction

As one of the indispensable parts of the transmission system in rotating machinery, the prognostic and health management (PHM) for gears has attracted increasing attention from academic and industrial fields in recent years [1–4]. Predicting the remaining useful life (RUL) of equipment in the field of PHM is difficult due to the complexity and severity of working conditions. If the gear life cannot be accurately estimated, a developing gear fault may result in transmission system failure or even a catastrophic event [5–8]. Therefore, utilizing run-to-failure datasets of equipment to conduct a precise and reliable RUL prediction is a practical obstacle.

In the field of RUL prediction, there are two primary approaches to predict the machine RUL. The first is the model-based approach [9–11], which employs physical laws and mathematical functions to describe the deterioration process of mechanical parts or systems with a well-established fault mechanism. The model-based method may lose effectiveness under harsh and uncontrollable working conditions when applied to a complex system. The second is the data-driven approach

[12–14] that uses artificial intelligence technologies to directly map the potential nonlinear relationship between machine degradation performance and measurement data. For RUL prediction of rotating machinery, the data-driven method requires minimal expert knowledge and is widely used. The current data-driven RUL prediction framework generally includes the following three steps: data input, feature extraction (HI construction), and RUL prediction (regression or time-series prediction). In the step of feature extraction (HI construction), the construction of health indicator (HI) plays a crucial role in the time-series-based RUL prediction method. A suitable HI can not only describe the machine's deterioration trend, but also improve RUL prediction accuracy. In recent years, several HI construction strategies for rotating machinery have been proposed. Guo et al. [15] proposed an unsupervised HI construction method based on a multi-scale convolutional auto-encoder. Xu et al. [16] presented a method for bearing HI construction using stacked auto-encoder with an exponential function. She et al. [17] constructed bearing HI using a sparse auto-encoder network with regularization. Qin et al. [18] developed a degradation-trend-constrained variational auto-encoder to build the

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machine's HI. According to the research of the literatures mentioned above, model training for these network-based HI construction methods requires a large amount of degradation data. However, it is difficult to collect sufficient run-to-failure data in the real world. In addition, the HI derived from the trained model cannot accurately reflect the degradation trend when the working condition varies. This is because the testing data may have different distributions with the training data. To address the issues mentioned above and realize the time-series-based life prediction under different domains, via a quadratic function-based multi-scale deep convolutional auto-encoder and maximum mean discrepancy, a gear HI transfer construction framework called TQFMDCAE is proposed for generating the cross-domain HIs. Based on the generated gear HIs, the RUL can be accurately predicted by the proposed RUL prediction model.

In the step of RUL prediction, pattern recognition-based life prediction techniques are favored by researchers. For instance, He et al. [19] proposed a semi-supervised generative adversarial network (GAN) for RUL estimation based on both failure and suspension histories. Xia et al. [20] presented a distance self-attention network (DSAN) for RUL prediction of aero-engine with parallel computing. Li et al. [21] developed a multi-scale deep convolutional neural network (MS-DCNN) to extract multi-scale features for machine RUL estimation. Liu et al. [22] designed a combined prediction model for multi-stage life prediction of engine. Xiang et al. [23] presented an automatic multi-differential learning deep neural network (AMLDNN) to utilize different levels of characteristic information for machine RUL prediction. Ma and Mao [24] created a convolution-based long short-term memory (CLSTM) network for estimating the RUL of a machine using vibration data.

Although data-driven approaches based on pattern recognition have been widely applied to RUL prediction, the predictive performance of these networks may be poor when the degeneration trend of the testing object differs significantly from that of the training object. In addition, abundant training datasets with labels are required for training these approaches. Obtaining sufficient training datasets could be time-consuming and expensive. In practice, the complete life-cycle data may be limited, even sometimes the life-cycle data may be incomplete. To address this problem, the time-series prediction method is developed to forecast the RUL in the absence of sufficient life-cycle data. Using deep neural networks, it can fit well with the degradation trend of the known degradation data and predict the unknown degradation data. The RUL is computed when the predicted degradation data reaches the predetermined failure threshold. Recent researches have investigated time-series prediction methods for RUL estimation. Lu et al. [25] proposed a rolling bearing life prediction framework based on deep adversarial learning. Xiang et al. [26] presented a new LSTM network variant with weight amplification (LSTMP-A) for predicting gear life. Chen et al. [27] created a gated adaptive hierarchical attention unit (GAHAU) for servo motor life prediction. Yan et al. [28] utilized an ordered neurons LSTM (ON-LSTM) for predicting the long-term gear life.

The pattern recognition-based RUL prediction method usually assumes that the data distributions of the training dataset and testing dataset are similar, so the trained RUL prediction model can be directly used to estimate the RUL of testing dataset. However, in real applications, the distribution shift of cross-domain data exists when the working condition changes. Thus, if the RUL prediction model trained via one dataset (source domain) is employed to predict the RUL of other datasets with different working conditions (target domains), the prediction performance will decrease significantly because of the varied data distributions. To effectively solve the foregoing issue, transfer learning has been introduced into the RUL prediction, and some transfer pattern recognition-based methods were presented to improve the accuracy of RUL prediction under the variable conditions. Mao et al. [29] presented a contractive denoising auto-encoder (CDAE) and transfer component analysis (TCA) for rolling bearing transfer RUL prediction. Miao and Yu [30] used a deep domain adaptive network (DDAN) to minimize the distribution discrepancy of features under various working conditions

for life estimation. Ding et al. [31] utilized two transfer learning strategies to transfer the bearing life prediction model from the source domain to the target domain. Fu et al. [32] introduced a deep residual LSTM with domain-invariance (DIDLSTM) to predict mechanical RULs under various working conditions. Li et al. [33] proposed an RUL prediction method based on a deep adversarial neural network to extract general prognostic knowledge for realizing rolling bearing data alignment. Kontar et al. [34] presented a multivariate Gaussian convolution process to transfer knowledge via sharing the latent functions between historical and in-service condition monitoring signals. Fallahdizch and Wang [35] developed a transfer learning framework that can improve the prognosis of an in-service unit via multivariate functional principal component analysis (MFPCA). In addition, they also proposed a data-level transfer learning framework to mine the effective information from various processes to improve the prognostic accuracy [36]. Although the transfer pattern recognition-based prediction method can perform the RUL prediction from source domain to target domain, the prediction accuracy may decrease greatly when the full life-cycle data (source domain data) is limited or even the life-cycle data (source domain data) is incomplete. Thus, it is necessary to improve the RUL prediction performance under various working conditions (various domains) by the time-series-based RUL prediction approach and feature transfer technology. The difference between the time-series-based and pattern recognition-based RUL transfer prediction methods is that the pattern recognition-based method that needs abundant data (source domain data) to train the model is an end-to-end framework, while the time-series-based RUL prediction method has two stages. In the first stage, the HI construction method is trained by the source-domain data and target-domain data, then the HI of target-domain data is generated through the trained model. In the second stage, the time-series-based life prediction model is trained by some known HI points, then the unknown HI points are predicted via the trained model step by step, finally the RUL can be obtained via the preset failure threshold. Compared with the pattern recognition-based RUL transfer prediction methods, the proposed approach can better achieve the transfer RUL prediction under the limited life-cycle samples.

Due to the vanishing gradient and gradient explosion of RNN, the LSTM network with a specific gate mechanism has been widely applied to the time-series prediction method for long-term RUL prediction. In LSTM network, the forget gate can determine how much previous information should be excluded at the current time step, and the input gate can absorb the relevant information into the cell state. For optimizing parameters more easily during the backpropagation, the sigmoid function is used to mimic the gates in practical application, whose output values are soft values between 0 and 1. However, by investigating the gate output of LSTM from a large number of experiments, we can find that the activation values of forget and input gates may not approach 0 or 1 and most of the activation values are around 0.5, which means that most activation values are ambiguous. This phenomenon does not conform to the logic of the gating mechanism, and it leads to the compromised long-term memory capacity of LSTM. Thus, the RUL prediction accuracy of deep learning-based method (e.g. LSTM-based method) is limited by the traditional gating mechanism, especially for long-term RUL prediction. In addition, the sequence information of LSTM is ignored, resulting in a chaotic arrangement of neurons. For example, the activation values of the forget gate in LSTM are a vector in which the values are not arranged in a particular order. The sequence of neurons is relevant to time-varying information in data-driven models, particularly time-series prediction models based on the HI (e.g., health status). To address the issues mentioned above, a novel network should be carefully designed to make the forget and input gate values closer to 0 or 1 and leverage the sequence information of neurons. For the long-term life prediction of gears, a new multi-hierarchical long-term memory augmented network (MLMA-Net) is proposed. First, a memory-augmented function is designed to push the output values of the forget and input gates closer to 0 or 1 so that the network can remember more

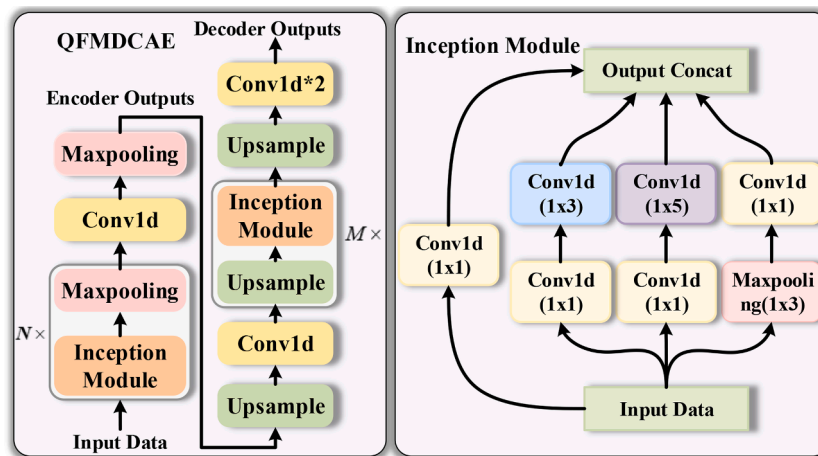


Fig. 1. The structure of the QFMDCAE network.

long-term data and update more current input data. In response to the output values of these two gates, a hierarchical attention gate is then developed. A multi-hierarchy mechanism divides the sequence information of this gate into three attention hierarchies (e.g., high attention hierarchy, middle attention hierarchy, and low attention hierarchy). The cell state is then updated by three hierarchies (e.g., high cell hierarchy, middle cell hierarchy, and low cell hierarchy) that are divided by a multi-hierarchical mechanism. Due to the memory augmentation function and multi-hierarchical mechanism described above, the proposed MLMA-Net is capable of making long-term RUL prediction. This paper develops a novel transfer RUL prediction methodology for gears using the TQFMDCAE and MLMA-Net. The experimental results indicate that the proposed RUL prediction approach outperforms other state-of-the-art RUL prediction methods, particularly for estimating the long-term RUL. The main contributions of this work are as follows.

- (1) In order to improve the efficacy of the activation values output by the gates, a memory-augmented function is first constructed to bring the values of forget and input gates close to 0 or 1. It increases long-term memory capacity and is consistent with the logic of the gate mechanism's design.
- (2) To make full use of the sequence information of LSTM, an adaptive multi-hierarchical mechanism is proposed for respectively dividing the sequence information of attention gate and

cell state into three attention hierarchies and three cell hierarchies.

- (3) Based on the proposed mechanism and the memory augmentation function, MLMA-Net is constructed, and the associated updating formulas are derived. The contrast experiments show its superiority over the state-of-the-art time-series predictive neural networks.
- (4) A HI transfer construction framework called TQFMDCAE is developed for generating the gear HIs under different working conditions. Using the cross-domain HIs and MLMA-Net, a novel gear life prediction method is proposed, which has stronger predictive ability than the existing advanced RUL prediction methods.

2. Theoretical background

This section describes the fundamental methods utilized in this study. TQFMDCAE is utilized to construct gear HIs cross domains. The MLMA-Net is proposed for RUL prediction with the constructed HIs based on LSTM.

2.1. Quadratic function-based multi-scale deep convolutional auto-encoder

Ref. [37] developed a quadratic function-based deep convolutional

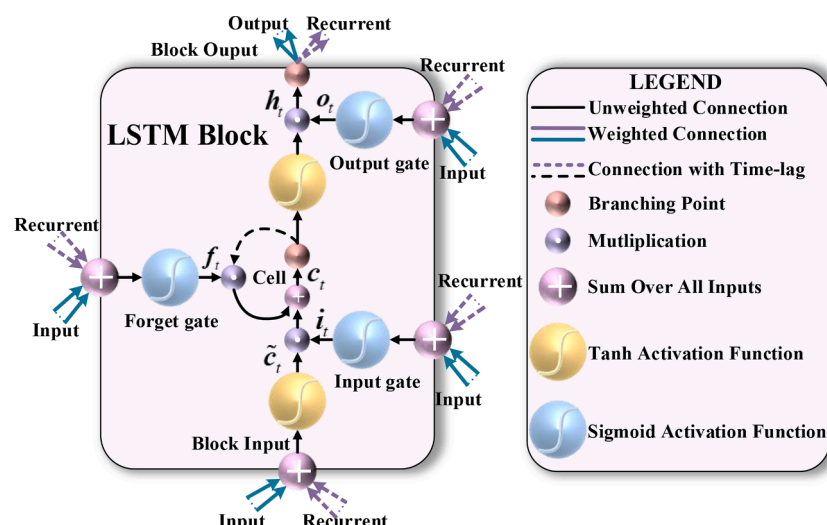


Fig. 2. The cell structure of the LSTM network.

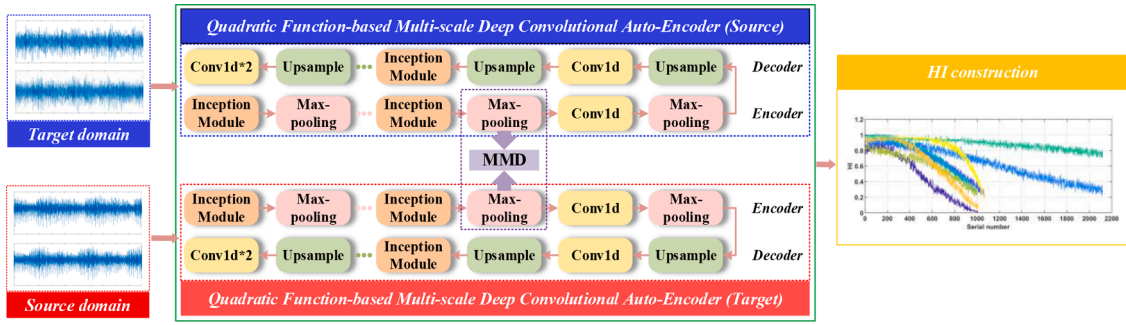


Fig. 3. The flowchart of gear HI transfer construction framework.

auto-encoder (QFDCAE) for constructing bearing HIs. The acquired HIs have superior monotonicity, temporal correlation, and robustness compared to other methods of HI construction. However, in each convolution layer of QFDCAE, only one convolution kernel is used for feature extraction from vibration data. This may weaken the model's ability to learn features and result in the loss of important feature data. Fortunately, GoogleNet's inception module provides a viable method for extracting multi-scale features via distinct convolution kernels. By incorporating the inception module into QFDCAE, the quadratic function-based multi-scale deep convolutional auto-encoder (QFMDCAE) network for gear HI construction is developed. Fig. 1 demonstrates the structure of QFMDCAE.

2.2. Long short-term memory

As one of the RNN's variants, LSTM can effectively handle gradient vanishing and gradient explosion. It introduces a novel internal state to convey linear recurrent information and nonlinear output information to the external state (e.g., hidden state). Meanwhile, three gates are used to control the flow of information. Fig. 2 illustrates the LSTM network's cell structure.

The hidden state $\mathbf{h}_t$ in the cell of LSTM is calculated as:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f), \quad (1)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i), \quad (2)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o), \quad (3)$$

$$\tilde{\mathbf{c}}_t = \varphi(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c), \quad (4)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \quad (5)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \varphi(\mathbf{c}_t), \quad (6)$$

where $\sigma(\cdot)$ and $\varphi(\cdot)$ are, respectively, the logistic sigmoid and hyperbolic tangent function; $\mathbf{x}_t$ is the input data at the current time; $\mathbf{h}_{t-1}$ represents the hidden state at the previous time; $\mathbf{f}_t$, $\mathbf{i}_t$, and $\mathbf{o}_t$ are the activation values of forget gate, input gate, and output gate, respectively. $\tilde{\mathbf{c}}_t$ represents the candidate state at the current time; $\mathbf{W}$, $\mathbf{U}$, and $\mathbf{b}$ are the learnable parameters that are updated via optimization algorithm in the training process. $\mathbf{c}_t$ is the cell state at the current time; $\mathbf{h}_t$ denotes the hidden state at the current time, and $\odot$ represents dot product.

3. The proposed transfer RUL prediction methodology

In the proposed transfer RUL prediction method, the cross-domain gear HIs are first obtained by TQFMDCAE; then, the MLMA-Net is constructed for time-series prediction with the cross-domain HI vector; and finally, the gear RUL prediction special steps are introduced.

3.1. Transfer construction of health indicator

The gear HI can be constructed using supervised or unsupervised methods, such as kernel principal component analysis (KPCA), deep neural network (DNN), auto-encoder (AE), etc., based on monitoring data (e.g., vibration signals). As one of the AE variants, QFDCAE is suitable for gear HI construction, as demonstrated in the literature. However, only the specific convolution kernel is utilized to extract abstract features in each convolution layer of QFDCAE. This may limit the feature extraction ability of the QFDCAE network. To handle this issue, the inception module is used to learn multi-scale feature representations, and QFMDCAE is proposed by incorporating it into the QFDCAE network. The proposed QFMDCAE network contains nine inception modules, with five inception modules in the encoder and four in the decoder.

In practice, there is not enough run-to-failure data to train the HI construction model, and the HI constructed from the trained model may not describe the degradation trend of gear under different working conditions. Moreover, with respect to the generated HIs under different working conditions, the failure threshold used for calculating the RUL cannot be uniformly determined. To achieve the gear HI construction under different working conditions and uniformly determine the failure threshold of RUL prediction, the knowledge transfer method is employed, and a gear HI transfer construction framework is built via QFMDCAE and MMD. This framework is named as TQFMDCAE, and its flowchart is shown in Fig. 3. Firstly, the raw vibration signals of source-domain and target domain are used as the input of QFMDCAE network. Secondly, the source-domain and target-domain features obtained via the penultimate max-pooling layer in the encoder of QFMDCAE are used to calculate MMD, then the feature distribution discrepancy is minimized via MMD. In TQFMDCAE, we use the inception module to implement the multi-scale feature extraction. After training the proposed TQFMDCAE, the target-domain data is input into the trained network, and the outputs of the encoder of the trained QFMDCAE are used as HI of gear. The obtained gear HI can reflect the degradation trend of gear, and the failure threshold of the gear HIs acquired under different working conditions can be uniformly determined, which is significant for RUL prediction in practical engineering.

3.2. Multi-hierarchical long-term memory augmented network

In LSTM, a gate mechanism is used to control the flow of information in the hidden state, and a memory unit is introduced to allow the LSTM network to capture key information at certain time steps. However, because the soft gates in LSTM do not align with the real gate mechanism, the information flow may not be better controlled (e.g., opening or closing). Additionally, the sequence information of neurons is not taken into account. Due to the issues mentioned above, LSTM's long-term memory capacity may be limited.

To simultaneously address these issues, we develop MLMA-Net, a new variant of LSTM. In this network, a novel memory-augmented

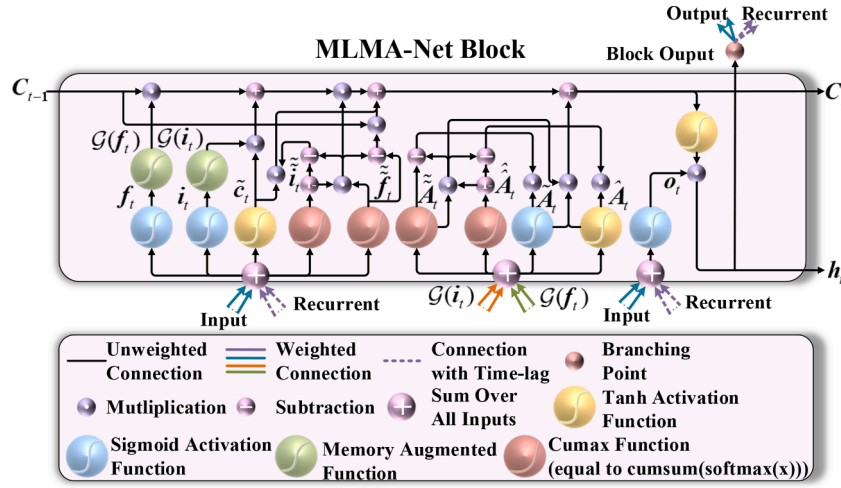


Fig. 4. The cell structure of MLMA-Net.

function is designed to bring the output values of the forget and input gates closer to 0 or 1, which is consistent with the gate mechanism's original intent. The sequence information of the attention gate and cell state is then separated using a multi-hierarchical mechanism. Lastly, the hidden state is updated using the hierarchical cell state and the attention gate's output value. The cell structure of MLMA-Net is shown in Fig. 4.

In LSTM, the sigmoid activation function imitates the gate, which can output soft values between 0 and 1. However, empirical evidence suggests that some gate values may not be close to 0 or 1. This contradicts the purpose of forget and input gates, which is to control whether information from previous time steps or current inputs is utilized. The direct method for bringing the values of forget and input gates closer to 0 or 1 is to modify the sigmoid function, which can bring more output values closer to 0 or 1. However, this is equivalent to readjusting the input value scale of gates and cannot guarantee output values that are closer to 0 or 1 for the learned gates.

To address this issue more effectively, we develop a novel memory-augmented function that modifies the output values of the gates directly. In addition, this function does not add any additional learning parameters. The memory-augmented function is computed utilizing the output values of forget or input gates:

$$G(x) = \begin{cases} 2x^2 & , 0 < x \leq 0.5 \\ -2x^2 + 4x - 1 & , 0.5 < x \leq 1, \end{cases} \quad (7)$$

where x represents the output value of forget or input gates. When x is 0.1, the output value of this function is 0.02, which is closer to 0. When x is 0.9, the output value of this function can reach 0.98, that is closer to 1. Moreover, there are no learning parameters in the proposed memory-augmented function, and this function can be used for any sigmoid gates in RNN-based network.

The memory-augmented function augments the capacity of long-term memory to some degree, but the output values of the memory-augmented function for forget and input gates are ignored, and the sequence information of this network is also disregarded. To address the issues mentioned above, an attention gate is designed to monitor the output values of the memory-augmented function for the two gates, which is calculated as follows:

$$\tilde{A}_t = \sigma(\mathbf{W}_A G(\mathbf{f}_t) + \mathbf{U}_A G(\mathbf{i}_t) + \mathbf{b}_A), \quad (8)$$

$$\hat{A}_t = \varphi(\mathbf{W}_A G(\mathbf{f}_t) + \mathbf{U}_A G(\mathbf{i}_t) + \mathbf{b}_A), \quad (9)$$

$$\mathbf{A}_t = \tilde{A}_t \odot \hat{A}_t, \quad (10)$$

where $\tilde{A}_t$ is the ratio of the output values of the memory augmentation function for the forget and input gates; $\hat{A}_t$ determines whether the output values of the memory augmentation function for the two gates are positive or negative; $\mathbf{A}_t$ denotes the attention distribution. To fully utilize the network's sequence information, a multi-hierarchical mechanism inspired by order neurons LSTM (ON-LSTM) [38] is used to mine sequence information. First, the attention gate's sequence information is divided into three attention hierarchies (e.g., high attention hierarchy, middle attention hierarchy, and low attention hierarchy). The master attention gates $\tilde{A}_t$ and $\hat{A}_t$ are defined as:

$$\begin{aligned} \mathbf{A}_{d_1} &= \text{softmax}(\mathbf{W}_{\sim} G(\mathbf{f}_t) + \mathbf{U}_{\sim} G(\mathbf{i}_t) + \mathbf{b}_{\sim}), \\ \mathbf{A}_{d_2} &= \text{softmax}(\mathbf{W}_{\sim}^A G(\mathbf{f}_t) + \mathbf{U}_{\sim}^A G(\mathbf{i}_t) + \mathbf{b}_{\sim}^A), \\ \tilde{\tilde{A}}_t &= \text{cumsum}(\mathbf{A}_{d_1}), \\ \hat{\hat{A}}_t &= 1 - \text{cumsum}(\mathbf{A}_{d_2}) \end{aligned} \quad (11)$$

Using (11), the high attention hierarchy η_{t2} , middle attention hierarchy η_{t1} , and low attention hierarchy η_{t3} are obtained by:

$$\eta_{t1} = \tilde{\tilde{A}}_t \odot \hat{\hat{A}}_t, \quad (12)$$

$$\eta_{t2} = \tilde{\tilde{A}}_t - \eta_{t1}, \quad (13)$$

$$\eta_{t3} = \hat{\hat{A}}_t - \eta_{t1}. \quad (14)$$

The Eq. (10) can be refined as

$$\mathbf{A}_t = \eta_{t3} \odot \hat{\hat{A}}_t + \eta_{t1} \odot (\tilde{\tilde{A}}_t \odot \hat{\hat{A}}_t) + \eta_{t2} \odot \tilde{\tilde{A}}_t. \quad (15)$$

In addition to the three hierarchies comprising the sequence information of the attention gate, the sequence information of the cell state is also divided into three distinct hierarchies (e.g., high cell hierarchy, middle cell hierarchy, and low cell hierarchy). The master forget gate $\tilde{f}_t$ and the master input gate $\tilde{i}_t$ are computed by the following equations:

$$\begin{aligned} \tilde{f}_t &= \text{softmax}(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f), \\ \tilde{i}_t &= \text{softmax}(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i), \\ \tilde{\tilde{f}}_t &= \text{cumsum}(\tilde{f}_t), \\ \tilde{\tilde{i}}_t &= 1 - \text{cumsum}(\tilde{i}_t) \end{aligned} \quad (16)$$

Next, the three different hierarchies are divided by:

$$\zeta_{t1} = \tilde{\tilde{f}}_t \odot \tilde{\tilde{i}}_t, \quad (17)$$

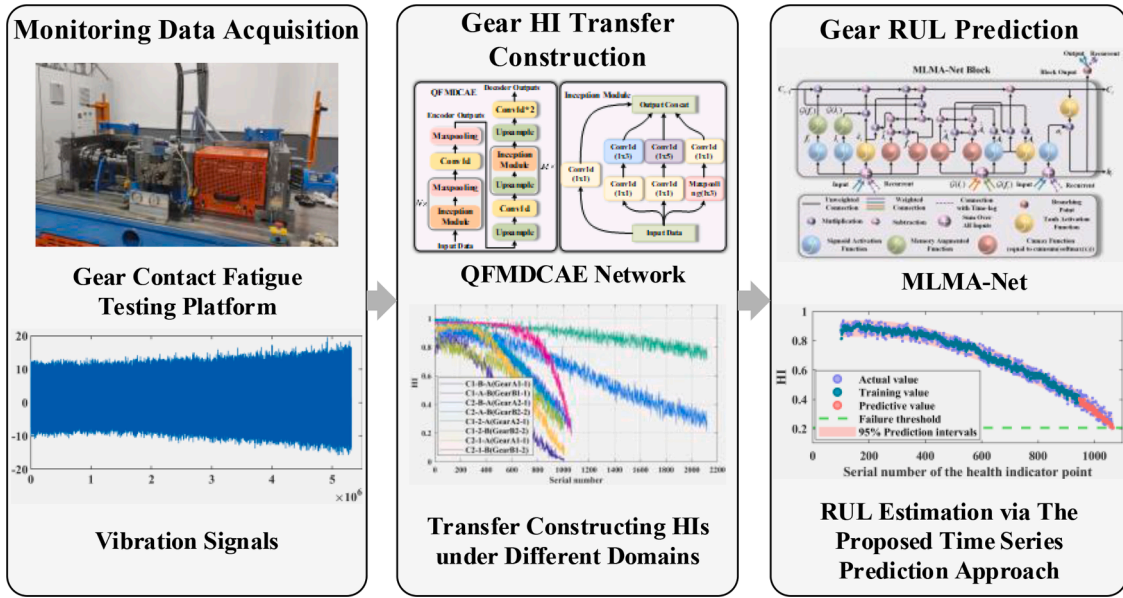


Fig. 5. The whole flowchart of the RUL prediction approach.

$$\xi_{t2} = \tilde{f}_t - \xi_{t1}, \quad (18)$$

$$\xi_{t3} = \tilde{l}_t - \xi_{t1}, \quad (19)$$

where ξ_{t1} denotes the middle cell hierarchy, ξ_{t2} denotes the high cell hierarchy, and ξ_{t3} denotes the low cell hierarchy.

The cell state $\mathbf{h}_t$ of MLMA-Net is updated by

$$\begin{aligned} \mathbf{c}'_t &= G(\mathbf{f}_t) \odot \mathbf{c}_{t-1} + G(\mathbf{i}_t) \odot \tilde{\mathbf{c}}_t, \\ \mathbf{c}_t &= \xi_{t2} \odot \mathbf{c}_{t-1} + \xi_{t1} \odot \mathbf{c}'_t + \xi_{t3} \odot \tilde{\mathbf{c}}_t + \mathbf{A}_t, \\ \mathbf{h}_t &= \mathbf{o}_t \odot \varphi(\mathbf{c}_t) \end{aligned} \quad (20)$$

Finally, the output of the MLMA-Net is obtained by

$$\begin{aligned} \mathbf{y}'_t &= \mathbf{W}_2(\text{relu}(\mathbf{W}_1 \mathbf{h}_t + \mathbf{b}_1)) + \mathbf{b}_2, \\ \mathbf{y}_t &= \mathbf{W}_3(\mathbf{y}'_t + \mathbf{h}_t) + \mathbf{b}_3, \end{aligned} \quad (21)$$

where $\mathbf{y}_t$ is the output of MLMA-Net; $\mathbf{W}$ and $\mathbf{b}$ are the learning parameters.

3.3. Transfer RUL prediction methodology for gears

The TQFMDCAE is applied to gear monitoring data to extract cross-domain gear HIs. For instance, building HIs with various gears under the same or different working conditions. Using the proposed MLMA-Net, the constructed cross-domain HIs are then used to predict gear RUL. Fig. 5 depicts the entire flowchart of the RUL prediction procedure.

The detailed steps are as follows:

- (1) Monitoring Data Acquisition: A series of run-to-failure vibration signals are obtained by accelerometers in different working conditions. The sampling frequency is f_s , sampling time, and sampling interval are T_p and T_q , respectively. The length of the life-cycle samples is K .
- (2) Gear HI Transfer Construction: The pre-processed monitoring data is applied as the input data of the QFMDCAE network. Then, the extracted abstract feature discrepancy of source and target domains is minimized as follows:

$$MMD^2(F_s, F_t) = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(F_s) - \frac{1}{n_t} \sum_{i=1}^{n_t} \phi(F_t) \right\|_{\mathcal{H}}^2, \quad (22)$$

where $\mathcal{H}$ denotes the RKHS, $\phi(\cdot)$ represents the nonlinear mapping from raw data space to RKHS. $F_s = \{f_k\}_{k=1}^{n_s}$ and $F_t = \{f_k\}_{k=1}^{n_t}$ are the extracted features of source and target domains.

Next, the HI of gear that comes from the target domain is constructed via the trained QFMDCAE. The HI vector is $\mathbf{h} = [h_1, h_2, h_3, \dots, h_K]^T$. The first k ($0 < k < K$) HI values are utilized to generate training data $\mathbf{z} = [h_1, h_2, h_3, \dots, h_k]$.

- (1) Gear RUL Prediction: Using the training data $\mathbf{z}$, a Hankel matrix is constructed as

$$\mathbf{M} = \begin{bmatrix} h_1 & h_2 & \dots & h_{k-i} \\ h_2 & h_3 & \dots & h_{k-i+1} \\ \vdots & \vdots & \ddots & \vdots \\ h_{i+1} & h_{i+2} & \dots & h_k \end{bmatrix} = \begin{bmatrix} M_1 \\ M_2 \\ \vdots \\ M_{i+1} \end{bmatrix}, \quad (23)$$

where i represents the length of the input data, and

$$\mathbf{M}_z = [h_z h_{z+1} \dots h_{k-z+1}] z = 1, 2, \dots, t + 1. \quad (24)$$

The matrix $\mathbf{M}' = [\mathbf{M}_1 \mathbf{M}_2 \dots \mathbf{M}_i]$ is applied as the input of MLMA-Net and the vector $\mathbf{M}_{i+1}$ is the label data in the training process of MLMA-Net. The loss function of the proposed method can be written as

$$L(\mathbf{M}_{i+1}, F(\mathbf{M}')) = \frac{1}{k-i} \sum_{v=1}^{k-i} [h_{i+v} - F(h_v, h_{v+1}, \dots, h_{v+i-1})]^2, \quad (25)$$

where $\mathcal{F}$ is the mapping function that is equivalent to the MLMA-Net. The real-time recurrent learning (RTRL) algorithm is used for updating parameters in MLMA-Net, and the vector $\tilde{\mathbf{M}}_{i+2}$ at the first prediction time (FPT) is obtained by the trained MLMA-Net, which is written as

$$\tilde{\mathbf{M}}_{i+2} = F(\mathbf{M}_2, \mathbf{M}_3, \dots, \mathbf{M}_{i+1}) \quad (26)$$

After repeating the above process, a series of output vectors are acquired. The j th output vector is formulated as

$$\begin{aligned} \tilde{\mathbf{M}}_j &= F(\mathbf{M}_{j-i}, \mathbf{M}_{j-i+1}, \dots, \mathbf{M}_{i+1}, \tilde{\mathbf{M}}_{i+2}, \dots, \\ &\tilde{\mathbf{M}}_{i+(j-i-2)}, \tilde{\mathbf{M}}_{i+(j-i-1)}) \end{aligned} \quad (27)$$

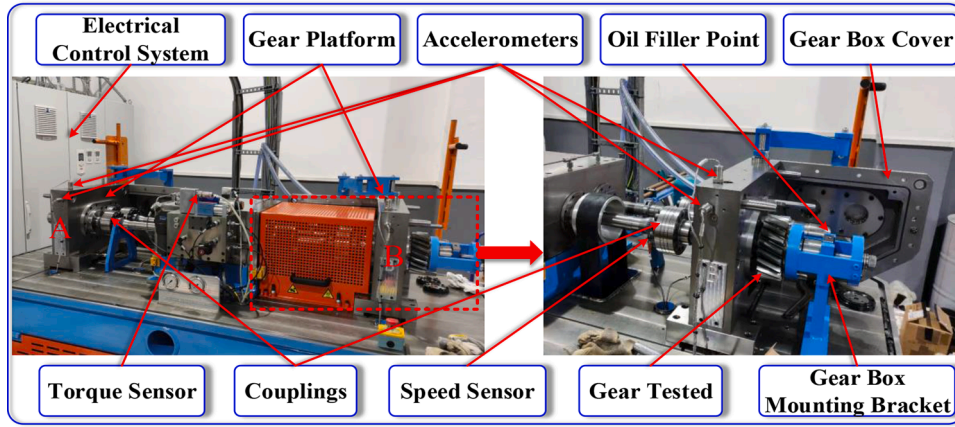


Fig. 6. Gear contact fatigue testing platform.

Table 1

The experimental data description.

Working condition	Condition1	Condition2
Rotating speed (rpm)	1100	800
Torque (Nm)	4400	4070
Gearbox A	GearA1-1	GearA2-1
	GearA1-2	GearA2-2
Gearbox B	GearB1-1	GearB2-1
	GearB1-2	GearB2-2

When the predicted value $\tilde{h}_{k-i+j-1}(\tilde{h}_{k-i+j-1} \in \tilde{M}_j)$ is smaller than the preset failure threshold, the gear RUL can be obtained via $(k - i + j - 1 - k) \times T_q$. Finally, the actual values are used for comparing with the predicted values that can verify the prediction performance of the MAML-Net.

4. Experimental results and analysis

To validate the prediction performance of MLMA-Net, the full life-cycle vibration signals are collected for gear RUL estimation under various working conditions. Then, via TQFMDCAE, the gear HIs for the same and different working conditions are constructed. The proposed method for predicting the RULs of gears is based on the constructed gear HIs. To demonstrate the long-term prediction capability of MLMA-Net, several conventional and cutting-edge RUL prediction methods are compared with the proposed method.

4.1. Experimental data description

On a gear contact fatigue testing platform, run-to-failure data is collected. As shown in Fig. 6, the testing platform comprises an electrical control system, a gear platform, four accelerometers, four couplings, a torque sensor, and a speed sensor. This testing platform has two gearboxes (e.g., gearbox A and gearbox B) with independent hydraulic lines for cooling the gears and bearings. Two pairs of gears can be tested concurrently under identical operating conditions.

In the experiment, the accelerometer was utilized to collect vibration signals from the A and B gearboxes. The sampling frequency was 25 kHz, and each sample contained 5000 points, so the sampling time was 0.2 s. The interval lasted 60 s. The gear test was complete when the amplitude of the vibration signal reached the predetermined value. There are eight datasets with two distinct working conditions, as shown in Table 1.

4.2. Transfer construction of gear health indicator

For the transfer construction of gear HI, the TQFMDCAE is used to

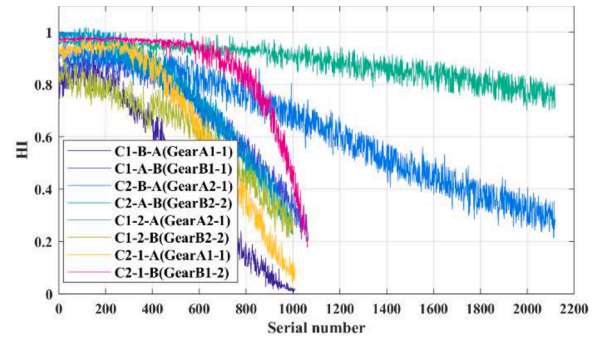


Fig. 7. The constructed gear HIs.

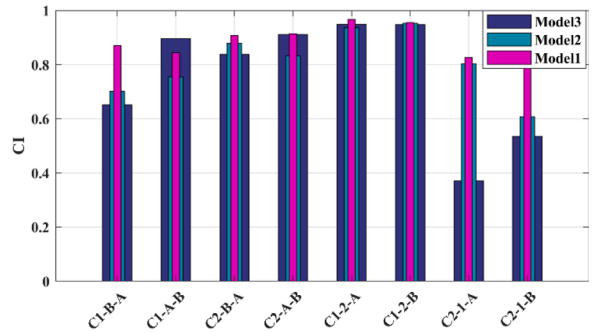


Fig. 8. The composite indexes of different gear HIs.

generate the cross-domain HIs from the vibration signals of gearbox A and gearbox B under the same or different operating conditions. The constructed gear HIs are depicted in Fig. 7. For the varied gear HIs, the learning rates are set to 0.003, 0.003, 0.0005, 0.003, 0.005, 0.003, 0.005 and 0.0005, respectively. Moreover, the Adam optimizer is applied to optimize the weight parameters in the TQFMDCAE model.

"C1-B-A(GearA1-1)" denotes the HI of gearbox A constructed by the labeled data of gearbox B under condition 1. The test data is the dataset gearA1-1. "C1-2-A(GearA2-1)" represents the HI of gearbox A constructed by the labeled data of gearbox A under condition 1. The data for testing is the gearA2-1 dataset. Similarly, the remainder of this figure can be explained by the descriptions provided above.

This figure demonstrates that the gear HI accurately represents gear degeneration. When the gear is in its early stages of operation, the degradation rate is low due to the absence of gear defects. After a period of time, the degradation rate has a greater value than that in the early

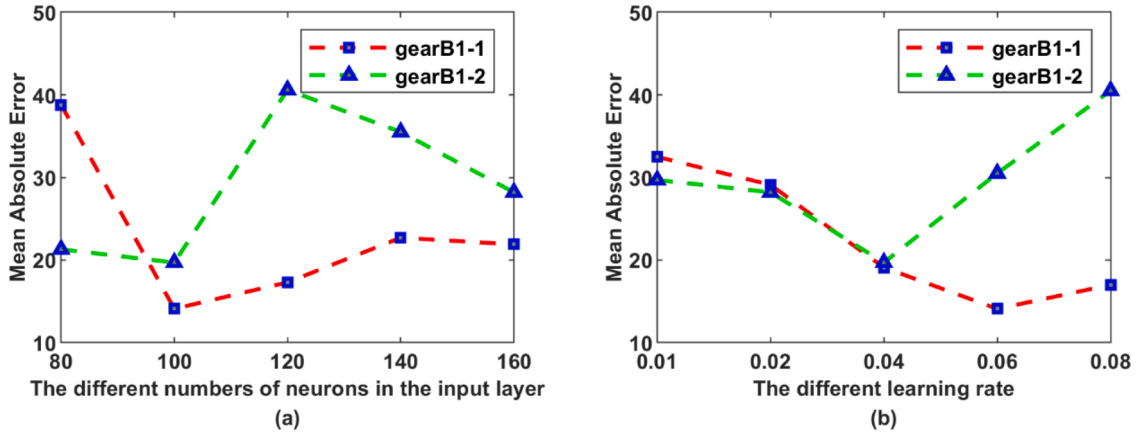


Fig. 9. Selecting hyper-parameters of MLMA-Net: (a) number of neurons in the input layer, (b) learning rate.

stage. This indicates that the gear is operating in an abnormal state and that some defects may develop.

To demonstrate the superiority of the TQFMDCAE (Model 1), the QFMDCAE (Model 2) and QFMDCAE with MMD (Model 3) methods are compared, and the composite index (CI) from Ref. [37] is used to assess the constructed HIs. The results are illustrated in Fig. 8. It can be seen from Fig. 8 that the MMD and inception module can improve the HI construction capacity of QFMDCAE. The constructed gear HIs are more appropriate for RUL gear prediction. Meanwhile, it can indicate from the result of model 3 that the bearing HI construction method proposed in Ref. [37] cannot be directly applied to construct gear HI under different domains.

4.3. RUL prediction and ablation experiment

To validate the proposed method's ability to predict RUL, the cross-domain HIs of the gearB1-1 and gearB1-2 datasets are used to estimate RUL. The total number of the two HIs is 1064, and the first 944 values are used to train MLMA-Net, followed by the prediction of the last 120 values by the optimized MLMA-Net. The actual RUL is 120 min, and via calculating the mean value of the minimum HI values obtained from

multiple experiments on gearbox B, the failure threshold is set as 0.206. In the proposed MLMA-Net, there are some hyper-parameters need to be selected, such as the number of neurons in the input layer and the learning rate. The hyper-parameters mentioned above will significantly influence the prediction performance of the MLMA-Net, so we employed the grid search method to select the hyper-parameters. First of all, the experiment on different number of neurons is repeated ten times, then the mean absolute error (MAE) is calculated to choose the hyper-parameter. The results obtained by different number of neurons are shown in Fig. 9(a). It can be seen that the optimized number of neurons in the input layer is 100 in both gearB1-1 and gearB1-2. We can also find that increasing the number of neurons in the input layer will decrease the prediction error when the number is less than 100. However, the prediction error will increase when the number of neurons in the input layer is greater than 100. This is because that more network parameters result in the overfitting for a limited-scale training dataset. After selecting the proper number of neurons in the input layer, the experiment on optimizing the learning rate is implemented. The results are depicted in Fig. 9(b). We can know from Fig. 9(b) that the optimized learning rates for gearB1-1 and gearB1-2 are 0.06 and 0.04, respectively. The large or small learning rate leads to the decline of prediction

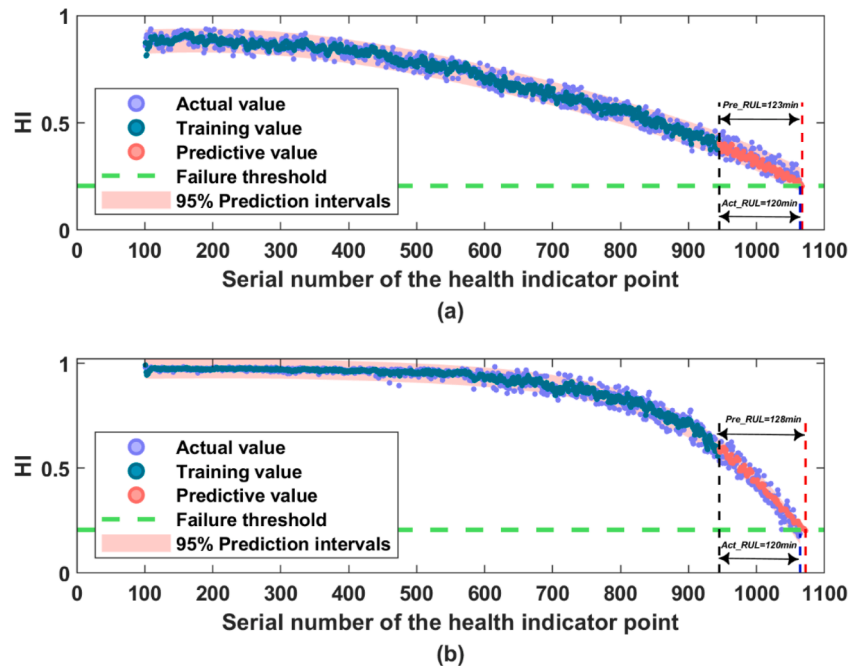
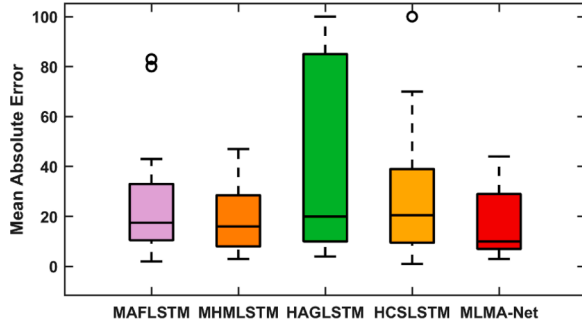


Fig. 10. Training and predicted HI values for 120 prediction values: (a) gearB1-1, (b) gearB1-2.

Table 2

The evaluation results of the ablation experiment.

	MAE	NRMSE	MAPE	RMSE	Score
MAFLSTM	25.4500	0.2600	21.21%	33.7068	0.3651
MHMLSTM	18.5000	0.1855	15.42%	21.7417	0.4673
HAGLSTM	39.1000	0.3538	32.58%	54.5170	0.2852
HCSLSTM	27.2000	0.2538	22.67%	36.0444	0.2923
MLMA-Net	16.9000	0.1746	14.08%	20.9141	0.5109

**Fig. 11.** Comparison of MAEs for different methods in the ablation experiment.

accuracy. This is mainly because that too large learning rate will cause model not to converge, while too small learning rate will lead to slow convergence or even non-convergence.

With the selected hyper-parameters, the construction of MLMA-Net is complete. The RULs of gearB1-1 and gearB1-2 are obtained after training the MLMA-Net with the specified hyper-parameters. To ensure that the predicted RULs are credible and reliable, the MLMA-Net is executed ten times on each gear HI, and the mean of the ten predicted RULs is used as the final RUL. For gearB1-1, the predicted RUL is 129 min; for gearB1-2, it is 111 min. Considering the most recent test, Fig. 10(a) and (b) depict the actual HI values, training HI values, predicted HI values, and 95% prediction intervals calculated using the MATLAB function "polyconf" for the two gear HIs.

The error between the training values and actual values is small, the trend of the predicted values is similar to that of the actual values, and all predicted values fall within the 95% predictive intervals. Due to the multi-hierarchical mechanism and memory-augmented function in MLMA-Net, we can conclude that the proposed RUL prediction method is effective for predicting gear RUL.

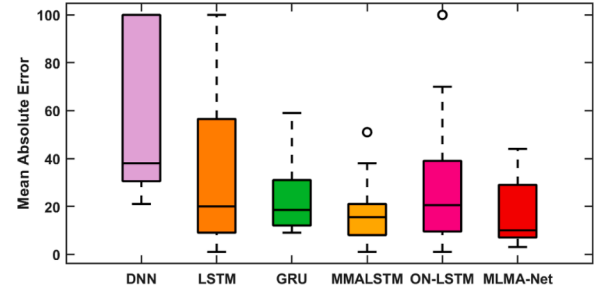
The memory-augmented function and multi-hierarchical mechanism in MLMA-Net are verified through an ablation experiment. Ablation experimentation employs the memory-augmented function-based LSTM (MAFLSTM), the LSTM network with multi-hierarchical mechanism (MHMLSTM), and the LSTM network only with hierarchy attention gate (HAGLSTM), and the LSTM network only with hierarchical cell state (HCSLSTM). These networks have the same hyper-parameters as MLMA-Net. MAE, normalized root mean square error (NRMSE), mean absolute percentage error (MAPE), root mean square error (RMSE), and score are used to quantify the prediction performance. In addition, these evaluation indicators can assess the uncertainty in prediction caused by the initialization of the network's weight parameter. Considering the predicted RULs of gearB1-1 and gearB1-2 synchronously, the values of these evaluation indexes are demonstrated in Table 2.

It is easily seen that the prediction performance of MLMA-Net is better than other models. According to these evaluation indexes, it has the highest score and the smallest prediction error. To provide the comprehensive evaluation results of RUL prediction, the boxplots of MAEs for various methods used in the ablation experiment are shown in Fig. 11. From the results of Fig. 11, we can find that the proposed MLMA-Net has shown a higher prediction performance on gear RUL prediction. The comparative results verify the role of memory-augmented function

Table 3

The comparison results of different methods.

	MAE	NRMSE	MAPE	RMSE	Score
DNN	61.5500	0.3903	51.29%	70.8544	0.0223
LSTM	34.1000	0.3212	28.42%	45.9315	0.3601
GRU	23.1000	0.2033	19.25%	27.4500	0.2827
MMALSTM	17.5000	0.1767	14.85%	21.5870	0.4626
ON-LSTM	27.2000	0.2538	22.67%	36.0444	0.2923
MLMA-Net	16.9000	0.1746	14.08%	20.9141	0.5109

**Fig. 12.** Comparison of MAEs for different methods.

and multi-hierarchical mechanism used in MLMA-Net. It can also know from the results of the evaluation indexes that the prediction errors of MAFLSTM and MHMLSTM are lower than those of HAGLSTM and HCSLSTM, and the scores of the two models are higher than HAGLSTM and HCSLSTM as well. This phenomenon indicates that the memory-augmented function and multi-hierarchical mechanism are beneficial to reduce the prediction error of gear RUL estimation. Consequently, the ablation results indicate that the MLMA-Net, which combines the memory-augmented function with the multi-hierarchical mechanism, is an efficient model for implementing RUL prediction.

4.4. Comparison with other methods

Conventional RUL prediction models, such as DNN, LSTM, GRU, MMALSTM [1], and ON-LSTM [38], are compared to the MLMA-Net to further demonstrate the remarkable prediction performance of the MLMA-Net. For the purpose of a fair comparison, the hyper-parameters of these models have been set to match those of MAML-Net, that is, the numbers of neurons in the input and output layers are 100 and 1, respectively. There are three hidden layers, and the numbers of neurons in these hidden layers are 20, 40, and 20, separately. The learning rates are set to 0.06 and 0.04 for the two cross-domain gear HIs, respectively. In Table 3, the comparative results are illustrated. In addition, the boxplot of MAE for these methods is shown in Fig. 12.

As shown in Table 3, the MLMA-Net has the highest score among these RUL prediction methods. In addition, the values of MAE, NRMSE, MAPE, and RMSE indicate that MLMA-Net has the lowest prediction error when compared to other methods. From Fig. 12, it can be known that the MLMA-Net has a stronger predictive ability than these conventional RUL prediction approaches. These results show that the proposed method has outstanding prediction performance because of the memory-augmented function and multi-hierarchical mechanism. In addition, we can also see that DNN, LSTM, and GRU networks have poor prediction performance among these methods, this is mainly because they do not apply any attention mechanism or hierarchical mechanism to improve the prediction accuracy of life prediction. MMALSTM and ON-LSTM have lower prediction errors than DNN and LSTM, since the utilized attention mechanism in MMALSTM and the designed hierarchical mechanism in ON-LSTM can learning more significant information from the input data to improve the prediction capacity of the life prediction methods. From these comparative results mentioned above, we can draw a conclusion that the memory-augmented function and the

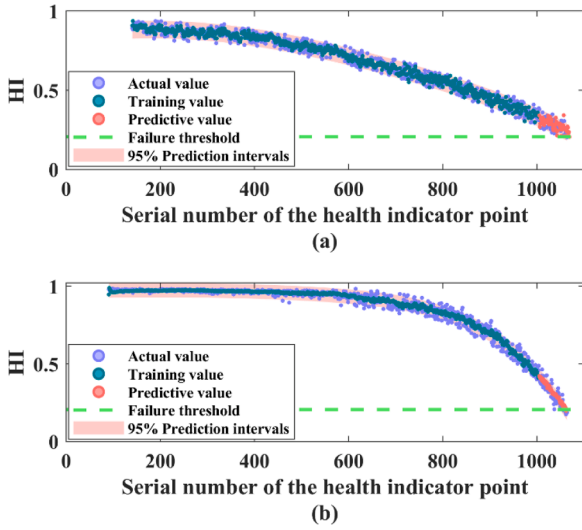


Fig. 13. Training and predicted HI values for 60 prediction values: (a) gearB1-1, (b) gearB1-2.

Table 4

The evaluation results for short-term prediction.

	MAE	NRMSE	MAPE	RMSE	Score
LSTM	28.9500	0.4633	48.25%	40.0056	0.1505
GRU	26.0000	0.6151	43.33%	40.7823	0.3746
MMALSTM	18.8000	0.4322	31.33%	28.1389	0.2869
ON-LSTM	28.6000	0.5716	47.67%	42.6392	0.3063
MLMA-Net	10.6000	0.2255	17.67%	13.8203	0.3951

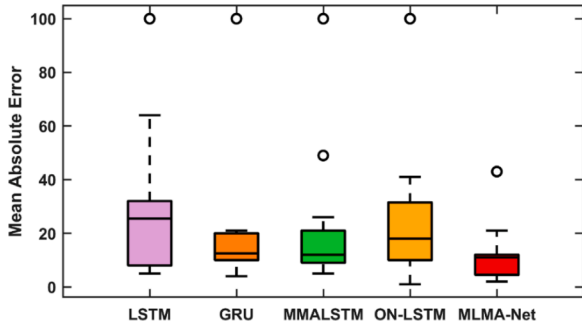


Fig. 14. Comparison of MAEs for different methods in the short-term RUL prediction.

multi-hierarchical mechanism extremely help to improve the prediction accuracy of MLMA-Net.

4.5. Short-term and long-term RUL prediction

This section verifies the short-term and long-term prediction performance of the proposed MLMA-Net. To demonstrate the short-term prediction performance of MLMA-Net, 60 values of the two HIs are predicted. The obtained prediction results are depicted in Fig. 13, and the evaluation results are presented in Table 4.

The predictive HI values are consistent with the trend of the actual HI values, as shown in Fig. 13. This indicates that the MLMA-Net is also capable of short-term prediction. Comparing the evaluation indexes of the presented model with those of the comparative methods reveals that the presented model has lower prediction error and a higher score, indicating that the MLMA-Net has greater short-term prediction ability than other RUL prediction methods. Considering the short-term RUL

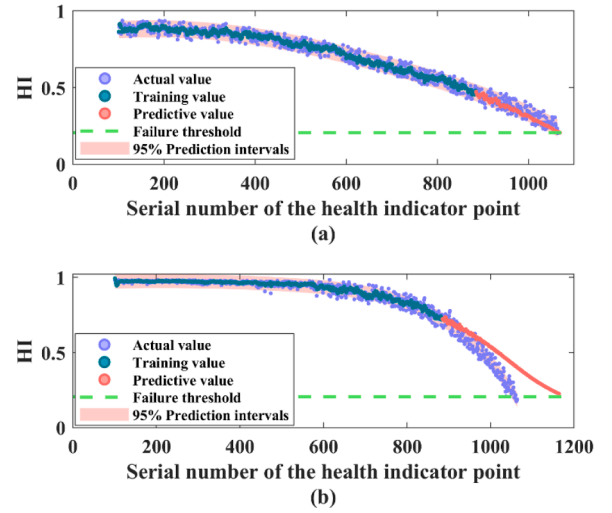


Fig. 15. Training and predicted HI values for 180 prediction values: (a) gearB1-1, (b) gearB1-2.

Table 5

The evaluation results for long-term prediction.

	MAE	NRMSE	MAPE	RMSE	Score
LSTM	74.4000	0.3226	41.33%	82.0573	0.0537
GRU	71.5000	0.3080	39.72%	77.4603	0.0291
MMALSTM	68.3000	0.3202	37.94%	78.6111	0.1284
ON-LSTM	73.3000	0.3049	40.72%	77.2263	0.0146
MLMA-Net	25.6000	0.1817	14.22%	35.1056	0.4784

prediction, the boxplots of MAEs obtained by various methods are illustrated in Fig. 14. This figure demonstrates that the proposed MLMA-Net has a superior prediction performance in the short-term prediction. From Table 4, it can be known that MMALSTM has lower prediction error than LSTM, GRU, and ON-LSTM because of the specific attention mechanism in MMALSTM. However, MMALSTM has poor prediction accuracy compare to MLMA-Net, this is mainly because the sequence information is not considered in MMALSTM network. Thus, we can conclude that the proposed MLMA-Net has better short-term prediction capacity than LSTM, GRU, MMALSTM, and ON-LSTM.

To demonstrate the long-term prediction performance of MLMA-Net, the first 884 HI values of gearB1-1 and gearB1-2 are used to train the proposed networks to predict the most recent 180 HI values. However, the RUL prediction of gearB1-2 is failed. This is mainly because many HI values of the training data are in the steady stage. These values will seriously prevent the MLMA-Net from learning degradation information in the training process. The predictive results are shown in Fig. 15. According to Fig. 15, the predicted HI values of gearB1-1 follow the same trend as the actual HI values, and all prediction values fall within the 95% predictive intervals. This indicates that the MLMA-Net can learn information about long-term degradation from the known HI values. Even though many stable HI values exist, Fig. 15(b) reveals that the proposed network can also mine some long-term degenerated information. Finally, we can conclude that the MLMA-Net has the capacity for long-term prediction and is appropriate for long-term RUL prediction.

Based on the two gear HIs, the LSTM, GRU, MMALSTM, and ON-LSTM are used for RUL prediction in order of comparison. Clearly, the RUL prediction of these methods for gearB1-2 also fails; only the RUL of gearB1-1 can be predicted using the networks mentioned above. Therefore, evaluation indexes for gearB1-1 are computed to quantify the prediction performance evaluation. The obtained results are shown in Table 5.

From Table 5, the proposed MLMA-Net has the best score that is

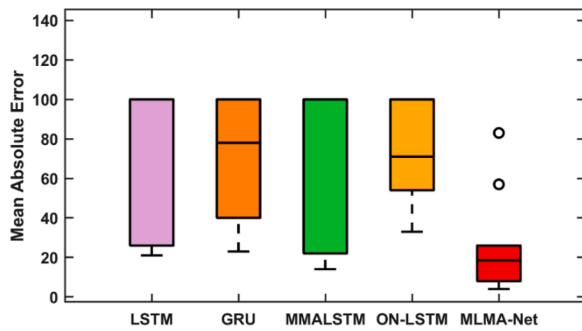


Fig. 16. Comparison of MAEs for different methods in the long-term RUL prediction.

much higher than that of other methods. Moreover, it has the smallest prediction error that can reflect from the calculated values of MAE, NRMSE, MAPE, and RMSE. Moreover, for the long-term RUL prediction, the MAEs obtained by various methods are shown in Fig. 16. From the result in this figure, we can easily find that MLMA-Net has the strongest predictive ability among these methods. We can also see from Table 5 that MMALSTM and ON-LSTM do not have better long-term prediction performance than LSTM and GRU, this is mainly because they do not simultaneously consider the attention mechanism and hierarchical mechanism, and the memory-augmented function is not applied in MMALSTM or ON-LSTM as well. The above phenomenon proves that the long-term prediction performance of MLMA-Net is obviously superior to other prediction methods. Therefore, it is more suitable for long-term RUL prediction.

5. Conclusions

To achieve the gear RUL prediction under different working conditions, a novel transfer RUL prediction methodology for gears was developed based on the proposed TQFMDCAE and MLMA-Net. The cross-domain HIs of gear were first constructed using TQFMDCAE with monitoring data. Then based on the gear HIs and the uniform failure threshold, MLMA-Net is proposed for predicting gear RULs under variable working conditions. In MLMA-Net, a memory-augmented function was created to bring the activation value of the gate closer to 0 or 1, which can increase the memory capacity of LSTM. Meanwhile, an adaptive hierarchical attention unit was designed to divide the sequence information of the attention gate into high attention hierarchy, middle attention hierarchy, and low attention hierarchy. Also, the sequence information of the cell state was separated into three cell hierarchies alongside the hidden state and input data. Due to its memory augmentation function and multi-hierarchical mechanism, the presented MLMA-Net has an exceptional capacity for prediction. With gear run-to-failure data with variable working conditions, the predictive performance of the developed RUL estimation method was demonstrated. Finally, comparisons were made between MLMA-Net and other RUL prediction methods. The comparative results indicate that MLMA-Net has reliable and robust prediction performance, especially for long-term RUL prediction. In future work, we will study how to suppress the negative transfer and perform more gear life-cycle tests for obtaining more vibration signals under other working conditions. Also, it is worth exploring how to integrate the time-series predictive network with transfer learning technologies. Additionally, a better memory augmentation function can be investigated to enhance the long-term memory capacity of LSTM-based neural network.

CRediT authorship contribution statement

Dingliang Chen: Writing – original draft, Formal analysis. **Yi Qin:** Supervision, Methodology. **Quan Qian:** Visualization, Data curation. **Yi**

Wang: Investigation, Validation. **Fuqiang Liu:** Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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